

Quadratic functions - Set 2

TMUA practice questions by topic. Specification MM1.3, M4.5, M4.11-M4.14. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the range of values of x for which $x^2 - 7x + 10 < 0$ and $x^2 - 4 > 0$.
- A $2 < x < 5$
B $-2 < x < 2$
C $x > 5$
D $x < -2$
E $2 < x < 5$ or $x < -2$
F no values
- 2 The expression $x^2 + 6x + c$ can be written as $(x + p)^2 + q$. Given that $q = -c$, find the value of c .
- A 9
B 3
C $-\frac{9}{2}$
D $\frac{9}{2}$
E 0
F -9
- 3 The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?
- A 3
B 1
C -1
D -3
E 5
F 2

- 4 The line $y = mx$ meets the curve $y = x^2 + 9$ at two distinct points. Find the complete set of values of m .
- A $m < -3$ or $m > 3$
B $m < -6$ or $m > 6$
C $-6 < m < 6$
D $-3 < m < 3$
E $m > 6$
F all real m
- 5 Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?
- A 8
B 6
C 5
D 7
E 3
F 4
- 6 The equation $x^2 + bx + c = 0$ has real coefficients. Which of the following is a **sufficient** condition for it to have two distinct real roots?
- A $b \neq 0$
B $b > 0$
C $c > 0$
D $b^2 \geq 4c$
E $c < 0$
- 7 The roots of $x^2 - 5x + 2 = 0$ are α and β . Find the value of $\alpha^2 + \beta^2$.
- A 25
B 29
C 23
D 27
E 21
F 17

- 8** The function $f(x) = x^2 - 6x + 13$ is defined for all real x . Which of the following statements is/are true?
- I** $f(x) > 0$ for all real x
 - II** The graph of $y = f(x)$ is symmetric about the line $x = 3$
 - III** The equation $f(x) = 4$ has exactly one real solution
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III
- 9** Find the sum of all real values of x satisfying $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.
- A** 0
B 9
C 3
D -3
E 6
F 2
- 10** The equation $x^2 - 2mx + m + 6 = 0$ has two distinct positive roots. Find the complete set of values of m .
- A** $-2 < m < 3$
B $m > 3$
C $-6 < m < -2$
D $m < -2$ or $m > 3$
E $m > 0$
F $m > -6$

- 11** The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I** If $ac < 0$, the equation has two distinct real roots.
- II** If $b = 0$, the equation has no real roots.
- III** If the equation has a repeated root, then $b^2 = 4ac$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 12** For how many integer values of k does $x^2 + kx + 12 = 0$ have integer roots?

- A** 8
- B** 4
- C** 2
- D** 12
- E** 3
- F** 6